

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$P(\text{buy} = y | x) = P(\text{age} \leq 30 | \text{yes}) + P(\text{inc} = \text{med} | \text{yes}) + P(\text{Stu} = \text{yes} | \text{yes}) + P(\text{Credit} = \text{Fair} | \text{yes})$$

$$= \frac{2}{9} \quad \frac{1}{9} \quad \frac{6}{9} \quad \frac{6}{9} \quad \frac{9}{14} \quad \left. \vphantom{\frac{2}{9}} \right\} \text{yes}$$

$$\frac{3}{5} \quad \frac{2}{5} \quad \frac{1}{5} \quad \frac{2}{5} \quad \frac{5}{14} \quad \left. \vphantom{\frac{3}{5}} \right\} \text{no}$$

Quiz 12

♀ age 20 student yes

$$P(\text{buy} = \text{yes} | x) = P(\text{age} \leq 30 | \text{yes}) \times P(\text{Student} = \text{yes} | \text{yes})$$

$$= \frac{2}{9} \times \frac{6}{9} = \frac{2}{9} \times \frac{1}{3} \times \frac{9}{14} = 0.048$$

$$P(\text{buy} = \text{no} | x) = P(\text{age} \leq 30 | \text{no}) \times P(\text{student} = \text{yes} | \text{no})$$

$$= \frac{3}{5} \times \frac{1}{5} \times \frac{5}{14} = 0.043$$